# DEBIASED ESTIMATORS IN HIGH-DIMENSIONAL REGRESSION: A REVIEW AND REPLICATION OF JAVANMARD AND MONTANARI (2014)

Benjamin Smith

Department of Statistical Sciences
University of Toronto
Toronto, Canada
benyamin.smith@mail.utoronto.ca

## ABSTRACT

High-dimensional statistical settings ($p \gg n$) pose fundamental challenges for classical inference, largely due to bias introduced by regularized estimators such as the LASSO. To address this, Javanmard and Montanari (2014) propose a debiased estimator that enables valid hypothesis testing and confidence interval construction. This report examines their debiased LASSO framework, which yields asymptotically normal estimators in high-dimensional settings. We present the key theoretical results underlying this approach, specifically, the construction of an optimized debiased estimator that restores asymptotic normality, which enables the computation of valid confidence intervals and $p$-values. To evaluate the claims of Javanmard and Montanari, a subset of the original simulation study and a re-examination of their real-data analysis are presented. Building on this baseline, we extend the empirical analysis to include the desparsified LASSO, a closely related method referenced but not implemented in the original study. The results demonstrate that while the debiased LASSO achieves reliable coverage and controls Type I error, the LASSO projection estimator can offer improved power in low-signal settings without compromising error rates. Our findings highlight a critical practical trade-off: while the LASSO projection estimator demonstrates superior statistical power in an idealized simulated low-signal setting, the estimation procedure employed by Javanmard and Montanari adapts more robustly to complex correlation networks, yielding superior precision and signal detection in real-world genomic data.

***Keywords*** Debiased LASSO · High-dimensional inference · Confidence intervals · Sparse regression

# 1 Introduction

Modern statistical problems are widely recognized as having become increasingly high-dimensional (Javanmard and Montanari, 2014). Examples such as genomics (Peng et al., 2010), neuroimaging (Yamashita et al., 2008), asset pricing (Kozak et al., 2019), signal processing (Lustig et al., 2007) and collaborative filtering[1](Koren et al., 2009) have led to settings in which the number of covariates exceeds the number of observations ($p > n$). LASSO regression (Tibshirani, 1996) has become a widely applicable and successful technique for addressing such problems. Estimators are derived by optimizing a regularized likelihood function, typically incorporating an $L_1$ penalty to induce sparsity. Because they are defined as the solution to a constrained optimization problem, LASSO estimators are inherently non-linear and do not admit an explicit, closed-form representation.

The non-linear and non-explicit nature of the estimators produced by LASSO estimates results in being unable to characterize the distribution the estimators it produces (Fu and Knight, 2000). This creates a challenge inference as one is unable to preform accepted procedures for characterizing uncertainty via confidence intervals and p-values (Wasserman, 2004; Lehmann and Romano, 2005). As of 2014[2] approaches to quantify statistical significance in high-dimensional parameter estimation settings has been less understood. Bühlmann (2013) and Zhang and Zhang (2014) introduced high-dimensional testing frameworks that utilize compatibility conditions to provide deterministic guarantees for significance $\alpha$ and power $1 - \beta$ under restricted eigenvalue or compatibility conditions (Bühlmann and van de Geer, 2011). These results were later strengthened by Javanmard and Montanari (2013), who established more refined lower bounds for the signal detection threshold. Belloni et al. (2013) and Belloni et al. (2014) consider inference for regression models with high-dimensional data. Meinshausen and Bühlmann (2010) and Minnier et al. (2011) studied resampling methods for hypotheses testing for these methods.

This report reviews and discusses the theoretical framework introduced by Javanmard and Montanari (2014) in their Journal of Machine Learning Research paper titled "Confidence Intervals and Hypothesis Testing for High-Dimensional Regression". Specifically, it examines the key lemmas and theorems underlying their approach to debiased inference for LASSO regression and presents a partial replication[3] of the original simulation study and an analysis of the data used in the original data analysis. The analysis compares their proposed method with the multisample splitting method (Meinshausen et al., 2009; Meinshausen and Bühlmann, 2010) and the ridge-type projection estimator (Bühlmann, 2013), as well as with alternative approaches implemented in the `hdi` R package (Ruben Dezeure et al., 2015). Furthermore, the comparison is extended to include the desparsified LASSO estimator (also known as the LASSO projection estimator, used interchangably) (van de Geer et al., 2014; Zhang and Zhang, 2014; Bühlmann and van de Geer, 2015), a closely related approach to debiased inference that was not formally evaluated (but referenced) in the original paper.

In Section 2, we outline the model setting and the de-biasing algorithm proposed by Javanmard and Montanari (2014), followed by an overview of technical prerequisites defined in Section 2.1. Section 2.2 presents the key theoretical results required for formal inference with the debiased estimator. Section 2.3 provides an overview of the primary applications derived from the theorems, including the construction of confidence intervals, hypothesis tests, and extensions to simultaneous inference and settings involving non-Gaussian noise. Section 3 assesses the empirical performance of Javanmard and Montanari (2014)'s debiased LASSO estimator through a partial replication of their simulation studies (Section 3.1) and a analysis of the riboflavin dataset (Bühlmann et al., 2014). By examining coverage probabilities, interval lengths, and Type I error rates, we compare the proposed method against competing approaches noted, but extend the analyses to include the LASSO projection estimator in order to highlight practical trade-offs. Section 4 concludes by combining the theoretical and empirical findings, discussing the practical trade-offs of debiased inference and its implications for high-dimensional statistical analysis in comparison with competing methods such as the LASSO projection estimator.

For reproducibility, the `R` code used to perform the simulations and data analysis is available at `https://github.com/phd-statistics/sta3000-report`.

# 2 Methodology

We work in a linear regression model setting where $n$ i.i.d. pairs $(Y_1, X_1), (Y_2, X_2), \ldots, (Y_n, X_n)$ are given with vectors $X_i \in \mathbb{R}^p$ and response variables defined as

$$Y_i = \langle \theta_0, X_i \rangle + W_i, \quad W_i \sim N(0, \sigma^2) \tag{1}$$

[1]Collaborative filtering is a core technique used in recommender systems that suggests items by finding patterns in user behaviour. It works under the assumption that individuals who agree in the past will agree in the future (Al-Ghuribi and Noah, 2021)

[2]This was the publication date of Javanmard and Montanari (2014) paper.

[3]Due to resource constraints, only a subset of the original simulation study is replicated.

Where $\theta_0 \in \mathbb{R}^p$ and $\langle \cdot, \cdot \rangle$ is the standard scalar product in $\mathbb{R}^p$. In matrix form, denote $Y = (Y_1, \ldots, Y_n)^\top$ and $\mathbf{X} \in \mathbb{R}^{n \times p}$. This gives

$$Y = \mathbf{X}\theta_0 + W, \quad W \sim N(0, \sigma^2 I_{n \times n}) \tag{2}$$

The goal is to estimate the unknown (but fixed) parameter vector $\theta_0 \in \mathbb{R}^p$. In a classical regression setting where $n \gg p$, the estimation method of choice is usually ordinary least squares, i.e. $\hat{\theta}^{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top Y$. By definition, $\hat{\theta}^{\text{OLS}}$ is Gaussian with mean $\theta_0$ and and covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$. This allows for typical inference techniques such as confidence intervals, hypothesis tests and construction of P-values.

In a high dimensional setting where the number of parameters exceeds the number of observations ($p > n$), the matrix $(X^\top X)$ is rank deficient and one is required to resort to biased estimators (Bühlmann and van de Geer, 2011). The LASSO (Chen and Donoho, 1995; Tibshirani, 1996) has become a widely used method that produces sparse estimates by incorporating an $\ell_1$ penalty into the objective function:

$$\hat{\theta}^n(Y, \mathbf{X}; \lambda) = \arg\min_{\theta \in \mathbb{R}^p} \left\{ \frac{1}{2n} \|Y - \mathbf{X}\theta\|_2^2 + \lambda \|\theta\|_1 \right\} \tag{3}$$

The $\ell_1$ penalty used in LASSO shrinks coefficients toward zero, thus introducing bias into the estimators. While this promotes sparsity, it also complicates statistical inference for the selected model as the resulting sampling distribution is difficult to characterize. Algorithm 1 shows the algorithm proposed by Javanmard and Montanari (2014) to construct the debiased estimator from the LASSO, given by Equation (5) as $\hat{\theta}^u = \hat{\theta}^n(\lambda) + \frac{1}{n} M \mathbf{X}^\top \big(Y - \mathbf{X}\hat{\theta}^n(\lambda)\big)$. The basic intuition is that the subgradient of the $\ell_1$ norm at the LASSO solution $\hat{\theta}^n$ is $\mathbf{X}^\top (Y - \mathbf{X}\hat{\theta}^n)/(n\lambda)$. By adding a term which is proportional to the subgradient, (5) compensates the bias introduced by the $\ell_1$ penalty in the LASSO.

**Algorithm 1** Unbiased estimator for $\theta_0$ in high-dimensional linear regression models

**Input:** Measurement vector $y$, design matrix $\mathbf{X}$, parameters $\lambda, \mu$.

**Output:** Unbiased estimator $\hat{\theta}^u$.

1: Let $\hat{\theta}^n = \hat{\theta}^n(y; \mathbf{X}; \lambda)$ be the LASSO estimator as per (3).

2: Set $\widehat{\Sigma} = (\mathbf{X}^\top \mathbf{X})/n$.

3: **for** $i = 1, 2, \ldots, p$ **do**

4: Let $m_i$ be a solution of the convex program:

$$\begin{aligned} \text{minimize} \quad & m^\top \widehat{\Sigma} m \\ \text{subject to} \quad & \|\widehat{\Sigma} m - e_i\|_\infty \leq \mu, \end{aligned} \tag{4}$$

where $e_i \in \mathbb{R}^p$ is the vector with one at the $i$-th position and zero everywhere else.

5: **end for**

6: Set $M = (m_1, \ldots, m_p)^\top$. If any of the above problems is not feasible, then set $M = I_{p \times p}$.

7: Define the estimator $\hat{\theta}^u$ as follows:

$$\hat{\theta}^u = \hat{\theta}^n(\lambda) + \frac{1}{n} M \mathbf{X}^\top \big(Y - \mathbf{X}\hat{\theta}^n(\lambda)\big) \tag{5}$$

It is claimed by Javanmard and Montanari (2014) that the estimator (5) guarantees near optimal confidence interval sizes and testing power. Section 2.1 presents preliminary definitions necessary for understanding the core theory presented in Javanmard and Montanari (2014). Section 2.2 presents the theorems which demonstrate that $\hat{\theta}^u$ is Gaussian and is an asymptotically unbiased estimator for $\theta_0$. Section 2.3 shows that the results from theorems in section 2.2 can be applied directly to derive confidence intervals (section 2.3.1), show near optimality of hypotheses tests generated (section 2.3.2),along with being generalized to derivation of simultaneous confidence intervals (Section 2.3.3) and the validity of the estimator $\hat{\theta}^u$ in non-Gaussian noise settings (Section 2.3.4).

### 2.1 Prerequisites

In this section, we will present basic definitions which will be used later in this paper.

**Definition 1** (True Support Set of $\theta_0$)**.** *We denote by $S \equiv supp(\theta_0)$ the support of $\theta_0 \in \mathbb{R}^p$, defined as:*

$$supp(\theta_0) \equiv \{i \in [p] : \theta_{0,i} \neq 0\} \tag{6}$$

*Where $[p] = \{1, \ldots, p\}$ is the index set of all parameters.*

In words, $\text{supp}(\theta_0)$ is the set of non-zero sparse parameters in the model (1). We further denote $s_0 \equiv |S|$ to be the number of elements in $\text{supp}(\theta_0)$.

**Definition 2** (Sub-Gaussian Norm)**.** *The sub-Gaussian norm of a random variable $X$, denoted by $\|X\|_{\psi_2}$ is defined as*

$$\|X\|_{\psi_2} = \sup_{q \geq 1} q^{-1/2} (\mathbb{E}|X|^q)^{1/q}$$

*For a random vector $X \in \mathbb{R}^n$, its sub-Gaussian norm is defined as*

$$\|X\|_{\psi_2} = \sup_{x \in \mathbb{S}^{n-1}} \|\langle X, x \rangle\|_{\psi_2}$$

*Where $\mathbb{S}^{n-1}$ denotes the unit sphere in $\mathbb{R}^n$*

The next two definitions provide the necessary foundation for the theorems in Section 2.2, which establish the unbiasedness of $\hat{\theta}^u$.

**Definition 3** (Compatibility Constant)**.** *Given a symmetric matrix $\hat{\Sigma} \in \mathbb{R}^{p \times p}$ and a set $S \subseteq [p]$, the corresponding compatibility constant is defined as*

$$\phi^2(\hat{\Sigma}, S) \equiv \min_{\theta \in \mathbb{R}^p} \left\{ \frac{|S| \langle \theta, \Sigma \hat{\theta} \rangle}{\|\theta_S\|_1^2} : \theta \in \mathbb{R}^p, \|\theta_{S^c}\|_1 \leq 3 \|\theta_S\|_1 \right\} \tag{7}$$

*We say that $\hat{\Sigma} \in \mathbb{R}^{p \times p}$ satisfies the compatibility condition for $S \subseteq [p]$ with constant $\phi_0$ if $\phi(\hat{\Sigma}, S) \geq \phi_0$. We say it holds for the design matrix $\mathbf{X}$ if it holds for $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$.*

**Definition 4** (Generalized Coherence Parameter)**.** *Given the pair $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $M \in \mathbb{R}^{p \times p}$. Let $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$ denote the associated sample covariance. The generalized coherence parameter of $\mathbf{X}, M$ is:*

$$\mu_*(\mathbf{X}; M) \equiv |M\hat{\Sigma} - I|_\infty = \max_{i \geq 1, j \leq p} |(M\hat{\Sigma})_{ij} - I_{ij}| = \max \left\{ \max_i |(M\hat{\Sigma})_{ii} - 1|, \max_{i \neq j} |(M\hat{\Sigma})_{ij}| \right\} \tag{8}$$

*We denote the minimum (generalized) coherence of $\mathbf{X}$ as $\mu_{\min}(\mathbf{X}) = \min_{M \in \mathbb{R}^{p \times p}} \mu_*(\mathbf{X}; M)$ and any minimizer of $\mu_*(\mathbf{X}; M)$ as $M_{min}(X)$.*

Note that the minimum coherence can be computed efficiently since $M \mapsto \mu_*(\mathbf{X}; M)$ is a convex function.

**Remark 5.** *If we assume that the columns of of $\mathbf{X}$ are normalized to have the $\ell_2$ norm equal to $\sqrt{n}$ (i.e., $\|\mathbf{X}e_i\| = \sqrt{n}$ for all $i \in [p]$), and $M = I$, then $(M\hat{\Sigma} - I)_{i,i} = 0$ and $|M\hat{\Sigma} - I|_\infty = \max_{i \neq j} |(\hat{\Sigma})_{i,j}|$. In other words, $\mu(\mathbf{X}; I)$ is the maximum normalized scalar product between distinct columns of $\mathbf{X}$. i.e.*

$$\mu_*(\mathbf{X}; I) = \frac{1}{n} \max_{i \neq j} |\langle \mathbf{X}e_i, \mathbf{X}e_j \rangle| \tag{9}$$

*Then the quantity (9) is known as the coherence parameter of the matrix $\mathbf{X}/\sqrt{n}$.*

Assuming for the sake of simplicity that the columns of $\mathbf{X}$ are normalized so that $\|\mathbf{X}e_i\|_2 = \sqrt{n}$, a small value of the coherence parameter $\mu_*(\mathbf{X}; I)$ means that the columns of $\mathbf{X}$ are roughly orthogonal. So $\mu_*(\mathbf{X}; I) = 0$ if and only if $\mathbf{X}/\sqrt{n}$ is column orthogonal. However, the generalized coherence parameter $\mu_*(\mathbf{X}; M)$ can be much smaller (i.e. it can remain small, potentially as small as order $O(\sqrt{(\log p)/n})$, even when the columns of $\mathbf{X}$ are significantly correlated).

Armed with these definitions, we now present the central theorems that establish the asymptotic normality of the debiased estimator $\hat{\theta}^u$.

### 2.2 A Debiased Estimator for $\theta_0$

Having established the generalized coherence $\mu_*$, we now characterize the distributional properties of the debiased estimator. The following theorems provide the conditions under which $\hat{\theta}^u$ is asymptotically normal, which will allow for classical inferential procedures such as confidence intervals and hypotheses testing. In order to clarify the distributional properties of $\hat{\theta}^u$, Javanmard and Montanari (2014) consider a general debiasing procedure that makes use of an arbitrary debiasing matrix $M \in \mathbb{R}^{p \times p}$ and define the *generalized debiased estimator* $\hat{\theta}^*$ as:

$$\hat{\theta}^*(Y, \mathbf{X}; M, \lambda) = \hat{\theta}^n(\lambda) + \frac{1}{n} M \mathbf{X}^\top (Y - \mathbf{X}\hat{\theta}^n(\lambda)) \tag{10}$$

**Theorem 6** (Error Decomposition around $\hat{\theta}^*$)**.** *Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be any (determinisitic) design matrix and $\hat{\theta}^* = \hat{\theta}^*(Y, \mathbf{X}; M, \lambda)$ be the generalized debiased estimator as per Equation (10).*

1. *Setting $Z = M\mathbf{X}^\top W/\sqrt{n}$ gives*

$$\sqrt{n}(\hat{\theta}^* - \theta_0) = Z + \Delta, \quad Z \sim N(0, \sigma^2 M\hat{\Sigma}M^\top), \quad \Delta = \sqrt{n}(M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n) \tag{11}$$

2. *Further, assume that $\mathbb{X}$ satisfies the compatibility condition for the set $S = supp(\theta_0), |S| \le s_0$, with constant $\phi_0$ and has a generalized coherence parameter $\mu_* = \mu_*(\mathbf{X}; M)$ and let $K \equiv \max_{i\in[p]} \hat{\Sigma}_{i,i}$. Then letting $\lambda = \sigma\sqrt{(c^2 \log p)/n}$ gives*

$$\mathbb{P}\left(\|\Delta\|_\infty \ge \frac{4c\mu_*\sigma s_0}{\phi_0^2}\sqrt{\log p}\right) \le 2p^{-c_0}, \quad c_0 = \frac{c^2}{32K} - 1 \tag{12}$$

*Note, if $M = M_{min}(X)$ minimizes $\mu_*(\mathbf{X}, M) = |M\hat{\Sigma} - I|_\infty$, then $\mu_*$ can be replaced by $\mu_{min}(X)$ in Equation (12).*

Theorem 6 decomposes the estimation error of the generalized debiased estimator $(\hat{\theta}^* - \theta_0)$ into a zero mean Gaussian term $Z/\sqrt{n}$ and a bias term $\Delta/\sqrt{n}$ whose maximum entry is bounded in probability per Equation 12.

*Proof.*

1. By substituting $Y = X\theta_0 + W$ into Equation (10), we can show that the estimation error $(\hat{\theta}^* - \theta_0)$ can be decomposed into the zero mean Gaussian term $Z/\sqrt{n}$ and the bias term $\Delta/\sqrt{n}$.

$$\begin{aligned}
\hat{\theta}^* &= \hat{\theta}^n + \frac{1}{n}M\mathbf{X}^\top(Y - X\hat{\theta}^n) \\
&= \hat{\theta}^n + \frac{1}{n}M\mathbf{X}^\top(\mathbf{X}\theta_0 + W - \mathbf{X}\hat{\theta}^n) \\
&= \hat{\theta}^n + \frac{1}{n}M\mathbf{X}^\top\mathbf{X}(\theta_0 - \hat{\theta}^n) + \frac{1}{n}M\mathbf{X}^\top W \\
&= \hat{\theta}^n + \frac{1}{n}M\mathbf{X}^\top\mathbf{X}(\theta_0 - \hat{\theta}^n) + \frac{1}{n}M\mathbf{X}^\top W - (\theta_0 - \hat{\theta}^n) + (\theta_0 - \hat{\theta}^n) \\
&= \theta_0 + \left(\frac{1}{n}M\mathbf{X}^\top\mathbf{X} - I\right)(\theta_0 - \hat{\theta}^n) + \frac{1}{n}M\mathbf{X}^\top W \\
&= \theta_0 + (M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n) + \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{n}}M\mathbf{X}^\top W \\
&= \theta_0 + \frac{1}{\sqrt{n}}\Delta + \frac{1}{\sqrt{n}}Z \\
&= \theta_0 + \frac{1}{\sqrt{n}}Z + \frac{1}{\sqrt{n}}\Delta \\
\iff (\hat{\theta}^* - \theta_0) &= +\frac{1}{\sqrt{n}}Z + \frac{1}{\sqrt{n}}\Delta
\end{aligned}$$

2. Noting that

$$\begin{aligned}
\|\Delta\|_\infty &= \|\sqrt{n}(M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n)\|_\infty \\
\left(\substack{\text{Triangle}\\ \text{Inequality}}\right) &\le \sqrt{n}\|M\hat{\Sigma} - I\|_\infty\|\theta_0 - \hat{\theta}^n\|_1 \\
&= \sqrt{n}\mu_*(\mathbf{X}; M)\|\theta_0 - \hat{\theta}^n\|_1 \\
&= \sqrt{n}\mu_*(\mathbf{X}; M)\|\hat{\theta}^n - \theta_0\|_1
\end{aligned}$$

By Bühlmann and van de Geer (2011, Theorem 6.1, Lemma 6.2),for any $\lambda \ge 4\sigma\sqrt{2K\log(pe^{t^2/2})/n}$

$$\mathbb{P}\left(\|\hat{\theta}^n - \theta_0\|_1 \ge \frac{4\lambda s_0}{\phi_0^2}\right) \le 2e^{-t^2/2} \tag{13}$$

So

$$\mathbb{P}\left(\|\Delta\|_\infty \geq \frac{4\lambda\mu_* s_0\sqrt{n}}{\phi_0^2}\right) \leq 2e^{-t^2/2} \tag{14}$$

The claim follows by selecting $t$ such that $e^{t^2/2} = p^{-c_0}$. Setting $\lambda = \sigma\sqrt{(K^2\log p)/n}$

$$\mathbb{P}\left(\|\Delta\|_\infty \geq \frac{4cs_0}{\phi_0^2}\sqrt{\log p}\right) \leq 2p^{-c_0}$$

□

The next theorem establishes that for a natural probabilistic model of the design matrix $\mathbf{X}$, the (constant) bound of the compatibility constant $\phi_0$ and the generalized coherence parameter $\mu_*(\mathbf{X}; M)$ can be bounded with probability converging rapidly to one as $n, p \to \infty$.

**Theorem 7** (Technicality)**.** *Let $\Sigma \in \mathbb{R}^{p\times p}$ be such that $\sigma_{min}(\Sigma) \geq C_{min} > 0$ and $\sigma_{max} \leq C_{max} < \infty$ and $\max_{i\in[p]} \Sigma_{ii} \leq 1$. Assume $\mathbf{X}\Sigma^{-1/2}$ to have independent sub-Gaussian rows, with zero mean and sub-Gaussian norm $\|\Sigma^{-1/2}X_1\|_{\psi_2} = \kappa$, for some constant $\kappa \in (0, \infty)$.*

*(a) For $\phi_0$, $s_0$, $K \in \mathbb{R}_{>0}$, let $\mathcal{E}_n = \mathcal{E}_n(\phi_0, s_0, K)$ denote the event that the compatibility condition holds for $\hat{\Sigma} = (\mathbf{X}^\top\mathbf{X}/n)$ for all sets $S \subseteq [p]$, $|S| \leq s_0$ with constant $\phi_0 > 0$ and that $\max_{i\in[p]} \Sigma_{i,i} \leq K$. i.e.*

$$\mathcal{E}_n(\phi_0, s_0, K) \equiv \left\{\mathbf{X} \in \mathbb{R}^{n\times p} : \min_{S:|S|\leq s_0} \phi(\hat{\Sigma}, S) \geq \phi_0, \quad \max_{i\in[p]} \hat{\Sigma}_{i,i} \leq K, \quad \hat{\Sigma} = (\mathbf{X}^\top\mathbf{X}/n)\right\} \tag{15}$$

*Then there exists $x_* \leq 2000$ such that the following happens. If $n \geq \nu_0 s_0 \log(p/s_0)$, $\nu_0 \equiv 5 \times 10^4 c_*(C_{max}/C_{min})^2\kappa^4$, $\phi_0 = \sqrt{C_{min}}/2$ and $K \geq 1 + 20\kappa^2\sqrt{(\log p)/n}$, then*

$$\mathbb{P}(\mathbf{X} \in \mathcal{E}_n(\phi_0, s_0, K)) \geq 1 - 4e^{-c_1 n}, \quad c_1 \equiv \frac{1}{4c_*\kappa^4} \tag{16}$$

*(b) For $a > 0$, let $\mathcal{G}_n = \mathcal{G}_n(a)$ be the event that the convex optimization problem in Algorithm 1 (i.e. (4)) is feasible for $\mu = a\sqrt{(\log p)/n}$. i.e.*

$$\mathcal{G}_n(a) \equiv \left\{\mathbf{X} \in \mathbb{R}^{n\times p} : \mu_{min}(\mathbf{X}) < a\sqrt{\frac{\log p}{n}}\right\} \tag{17}$$

*Then for all $n \geq a^2 C_{min} \log p/(4e^2 C_{\max}\kappa^4)$*

$$\mathbb{P}(X \in \mathcal{G}_n(a)) \geq 1 - 2p^{-c_2}, \quad c_2 \equiv \frac{a^2 C_{\min}}{24e^2\kappa^4 C_{\max}} - 2 \tag{18}$$

Theorem 7 shows for a natural probabilistic model of the design matrix $\mathbf{X} \in \mathbb{R}^{n\times p}$, both $\phi_0$ and $\mu_*(\mathbf{X}; M)$ can be bounded with probability that both converge rapidly to 1 as $n, p \to \infty$. Specifically, it guarantees that for sufficiently large $n$, the design matrix remains numerically stable enough to recover the true parameters while maintaining the low coherence required to filter out the shrinkage bias inherent in the LASSO. The proof for this theorem is omitted as it is a more nuanced technicality. For the proof, see the appendix of Javanmard and Montanari (2014).

Theorems 6 and 7 establish the necessary probabilistic guarantees for recovering asymptotically unbiased and Gaussian estimates of $\theta_0$. The following theorem builds on these results to characterize the limiting distribution of the debiased estimator $\hat{\theta}^u$ produced by Algorithm 1.

**Theorem 8** (Asymptotic Normality and Error Bounds for the Debiased LASSO Estimator $\hat{\theta}^u$)**.** *Consider the linear model (1) and the debiased LASSO estimator (5) defined in Algortihm 1 with $\mu = a\sqrt{(\log p)/n}$. Then setting $Z = M\mathbf{X}^\top W/\sqrt{n}$, we have*

$$\sqrt{n}(\hat{\theta}^u - \theta_0) = Z + \Delta, \quad Z|\mathbf{X} \sim N(0, \sigma^2 M\hat{\Sigma}M^\top), \quad \Delta = \sqrt{n}(M\hat{\Sigma} - I)(\theta_0 - \hat{\theta}^n) \tag{19}$$

*Further, under the assumptions of Theorem 7 for $n \geq \max(\nu_0 s_0 \log(p/s_0), \nu_1 \log p)$, $\nu_1 = \max(1600\kappa^4, a^2/(4e^2\kappa^4))$ and $\lambda = \sigma\sqrt{(c^2\log p)/n}$*

$$\mathbb{P}\left\{\|\Delta\|_\infty \geq \left(\frac{16ac\sigma}{C_{\min}}\right)\frac{s_0\log p}{\sqrt{n}}\right\} \leq 4e^{-c_1 n} + 4p^{-\min(\tilde{c}_0, c_2)} \tag{20}$$

*Where* $\tilde{c}_0 = (c^2/48) - 1$, $c_1 = \frac{1}{4c_*\kappa^4}$ *and* $c_2 = \frac{a^2 C_{\min}}{24e^2\kappa^4 C_{\max}}$.

*Finally the tail bound (20) holds for any choice* $M$ *that is a function of the design matrix and satistfies the feasibility condition of Equation (4) in Algorithm 1, i.e.* $|M\hat{\Sigma} - I|_\infty \leq \mu$.

Assuming $\sigma$, $C_{\min}$ of order 1, Theorem 8 establishes that for random designs, the maximum size of the bias term $\Delta_i$ over $i \in [p]$ is

$$\|\Delta\|_\infty = O\left(\frac{s_0 \log p}{\sqrt{n}}\right) \tag{21}$$

On the other hand, the 'noise term' $Z_i$ is roughly of order $\sqrt{[M\hat{\Sigma}M^\top]_{ii}}$

**Remark 9.** *Theorem 8 only requires that the support size satisfies* $s_0 = O(n/\log p)$. *If we further assume* $s_0 = o(\sqrt{n}/\log p)$ *then we have* $\|\Delta\|_\infty = o(1)$ *with high probability. Hence* $\hat{\theta}^u$ *is an asymptotically unbiased estimator for* $\theta_0$.

We will now present the proof for Theorem 8:

*Proof.* Denote $\|\Delta\|_0 \equiv \left(\frac{16ac\sigma}{C_{\min}}\right)\frac{s_0 \log p}{\sqrt{n}}$. We can then show that

$$\begin{aligned}
\mathbb{P}\left\{\|\Delta\|_\infty \geq \left(\frac{16ac\sigma}{C_{\min}}\right)\frac{s_0\log p}{\sqrt{n}}\right\} &= \mathbb{P}\left\{\|\Delta\|_\infty \geq \Delta_0\right\} \\
(*) \quad &\leq \mathbb{P}\Big(\left\{\|\Delta\|_\infty \geq \Delta_0\right\} \cap \mathcal{E}_n(\sqrt{C_{\min}}/2, s_0, 3/2) \cap \mathcal{G}_n(a)\Big) \\
&\quad + \mathbb{P}\left(\mathcal{E}_n^c(\sqrt{C_{\min}}/2, s_0, 3/2)\right) + \mathbb{P}(\mathcal{G}_n^c(a)) \\
(**) \quad &\leq \mathbb{P}\Big(\left\{\|\Delta\|_\infty \geq \Delta_0\right\} \cap \mathcal{E}_n(\sqrt{C_{\min}}/2, s_0, 3/2) \cap \mathcal{G}_n(a)\Big) \\
&\quad + 4e^{-c_1 n} + 2p^{-c_2}
\end{aligned}$$

Where the first equation $(*)$ comes from the law of total probability and $(**)$ comes from the probabilities of the compliments stated in Theorem 7. We note further that the bound from Theorem 7 (b) (i.e. $\mathbb{P}(\mathbf{X} \in \mathcal{E}_n(\phi_0, s_0, K)) \geq 1 - 4e^{-c_1 n}, \quad c_1 \equiv \frac{1}{4c_*\kappa^4}$) can be applied for $K = 3/2$ since under present assumptions $K \geq 1 + 20\kappa^2\sqrt{(\log p)/n} \iff 3/2 \geq 1 + 20\kappa^2\sqrt{(\log p)/n} \iff 1/2 \geq 20\kappa^2\sqrt{(\log p)/n}$.

$$\begin{aligned}
\mathbb{P}\Big(\left\{\|\Delta\|_\infty \geq \Delta_0\right\} \cap \mathcal{E}_n(\sqrt{C_{\min}}/2, s_0, 3/2) \cap \mathcal{G}_n(a)\Big) &\leq \sup_{\mathbf{X} \in \mathcal{E}_n(\sqrt{C_{\min}}/2, s_0, 3/2) \cap \mathcal{G}_n(a)} \mathbb{P}\left(\|\Delta\|_\infty \geq \Delta_0 \mid \mathbf{X}\right) \\
&\leq 2p^{-\tilde{c}_0}
\end{aligned}$$

Where the last inequality follows from Theorem 6. Thus

$$\mathbb{P}\left\{\|\Delta\|_\infty \geq \left(\frac{16ac\sigma}{C_{\min}}\right)\frac{s_0\log p}{\sqrt{n}}\right\} \leq 4e^{-c_1 n} + 2p^{-c_2} + 2p^{-\tilde{c}_0} \leq 4e^{-c_1 n} + 4p^{-\min(\tilde{c}_0, c_2)}$$

Where the last inequality follows by bounding the sum by the dominant rate □

Theorem 8 shows that asymptotic normality and error bounds of the generalized debiased estimator $\hat{\theta}^*$ can be applied to the debiased estimtor proposed by Javanmard and Montanari (2014) in Algorithm 1 $\hat{\theta}^u$ (See Equation (10)).

### 2.3 Statistical Inference

A direct application of Theorem 8 is to derive confidence intervals and statistical hypothesis tests for high dimensional models. The next Lemma will show how this is possible. Note, the sparisty assumption is assumed to be $s_0 = o(\sqrt{n}/\log p)$.

**Lemma 10** (Asymptotic Normality and Valid Inference for the Debiased LASSO)**.** *Consider a sequence of design matrices* $\mathbf{X} \in \mathbb{R}^{n \times p}$ *with dimension* $n \to \infty$, $p = p(n) \to \infty$ *satisfying the following assumptions for the constants* $C_{\min}$,$C_{\max}$, $\kappa \in (0, \infty)$ *independent of* $n$. *For each* $n, \Sigma \in \mathbb{R}^{p \times p}$ *is such that* $\sigma_{\min}(\Sigma) \geq C_{\min} > 0$ *and* $\sigma_{\max}(\Sigma) \leq C_{\max} < \infty$ *and* $\max_{i \in [p]} \Sigma_{ii} \leq 1$. *Assume* $\mathbf{X}\Sigma^{-1/2}$ *to have independent sub-Gaussian rows with zero*

*mean and sub-Gaussian norm* $\|\Sigma^{-1/2}X_1\|_{\psi_2} \leq \kappa$.

*Consider the linear model (1) and let* $\hat{\theta}^u$ *be the debaised LASSO estimate (5) derived in Algorithm 1 with* $\mu = a\sqrt{(\log p)/n}$ *and* $\lambda = \sigma\sqrt{(c^2 \log p)/n}$ *with* $a$ *and* $c$ *being large enough constants. Finally, let* $\hat{\sigma} = \hat{\sigma}(y, \mathbf{X})$ *be an estimator of the noise level satisfying for any* $\varepsilon > 0$

$$\lim_{n\to\infty} \sup_{\theta_0 \in \mathbb{R}^p; \|\theta_0\|_0 \leq s_0} \mathbb{P}\left(\left|\frac{\hat{\sigma}}{\sigma} - 1\right| \geq \varepsilon\right) = 0 \tag{22}$$

*If* $s_0 = o(\sqrt{n}/\log p)$ $(s_0 \geq 1)$*, then for all* $x \in \mathbb{R}$*, we have*

$$\lim_{n\to\infty} \sup_{\theta_0 \in \mathbb{R}^p; \|\theta_0\|_0 \leq s_0} \left|\mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x\right) - \Phi(x)\right| = 0 \tag{23}$$

*Where* $\Phi(\cdot)$ *denotes the cumulative distribution function (CDF) of a standard normal random variable.*

While the standard LASSO is an excellent tool for variable selection, its inherent bias makes it unsuitable for traditional statistical inference. Lemma 10 provides the necessary theoretical bridge, transforming $\hat{\theta}^u$ into a standardized statistic that converges to a standard normal distribution. This in turn shows that it is permissible to perform 'OLS-style' inference in settings where $p \gg n$

*Proof.* Under stated the stated assumptions in Lemma 10,we can prove

$$\lim_{n\to\infty} \sup \sup_{\|\theta_0\|_0 \leq s_0} \mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x\right) \leq \Phi(x) \tag{24}$$

By Theorem 8, we have

$$\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\sigma[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} = \frac{e_i^\top M\mathbf{X}^\top W}{\sigma[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} + \frac{\Delta_i}{\sigma[M\hat{\Sigma}M^\top]_{i,i}^{1/2}}$$

Let $V = \mathbf{X}M^\top e_i/(\sigma[M\hat{\Sigma}M^\top]_{ii}^{1/2})$ and $\tilde{Z} \equiv V^\top W$. Conditioned on $\mathbf{X}$ (and thus $V$), the independence of the noise $W \sim N(0, \sigma^2 I_n)$ implies:

$$\mathbb{P}(\tilde{Z} \leq x) = \mathbb{E}\left\{\mathbb{P}(V^\top W \leq x|V)\right\} = \mathbb{E}\left\{\Phi(x)|V\right\} = \Phi(x)$$

In order to prove Equation (24), fix $\varepsilon > 0$ and write

$$\begin{aligned}
\mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x\right) &= \mathbb{P}\left(\frac{\sigma}{\hat{\sigma}}\tilde{Z} + \frac{\Delta_i}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x\right) \\
&\leq \mathbb{P}\left(\frac{\sigma}{\hat{\sigma}}\tilde{Z} \leq x + \varepsilon\right) + \mathbb{P}\left(\frac{\Delta_i}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \geq \varepsilon\right) \\
&\leq \mathbb{P}\left(\tilde{Z} \leq x + 2\varepsilon + \varepsilon|x|\right) + \mathbb{P}\left(\frac{\Delta_i}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \geq \varepsilon\right) + \mathbb{P}\left(\left|\frac{\hat{\sigma}}{\sigma} - 1\right| \geq \varepsilon\right)
\end{aligned}$$

Taking the limit and using assuming that $\lim_{n\to\infty} \sup_{\theta_0 \in \mathbb{R}^p; \|\theta_0\|_0 \leq s_0} \mathbb{P}\left(\left|\frac{\hat{\sigma}}{\sigma} - 1\right| \geq \varepsilon\right) = 0$ we obtain:

$$\lim_{n\to\infty} \sup_{\theta_0 \in \mathbb{R}^p; \|\theta_0\|_0 \leq s_0} \mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x\right) \leq \Phi(x + 2\varepsilon + \varepsilon|x|) + \lim_{n\to\infty} \sup_{\theta_0 \in \mathbb{R}^p; \|\theta_0\|_0 \leq s_0} \mathbb{P}\left(\frac{\Delta_i}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \geq \varepsilon\right)$$

Since $\varepsilon > 0$ is arbitrary, showing that the limit vanishes requires bounding the absolute difference from both sides. By the properties of probability measures for any $\varepsilon > 0$, we have:

$$\left|\mathbb{P}\left(\frac{\sqrt{n}(\hat{\theta}_i^u - \theta_{0,i})}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \leq x\right) - \Phi(x)\right| \leq \mathbb{P}\left(\frac{|\Delta_i|}{\hat{\sigma}[M\hat{\Sigma}M^\top]_{i,i}^{1/2}} \geq \varepsilon\right) + \delta_\varepsilon(x) \tag{25}$$

where $\delta_\varepsilon(x)$ represents the modulus of continuity of $\Phi(\cdot)$, which vanishes as $\varepsilon \to 0$. By Theorem 8, the first term on the right-hand side converges to zero as $n \to \infty$. Since this holds for any $\varepsilon$, the supremum of the absolute difference over the sparsity class $s_0$ must likewise converge to zero. □

Lemma 10 is the bridge which facilitates standard inferential techniques such as confidence intervals and hypothesis testing for $\hat{\theta}^u$. Furthermore, this result is generalizable to simultaneous inference and remains robust under non-Gaussian noise distributions. These results are presented in subsections 2.3.1,2.3.2,2.3.3 and 2.3.4 without proof for brevity.

### 2.3.1 Confidence Intervals

From Lemma 10 it is quite straightforward to construct asymptotically valid confidence intervals. For $i \in [p]$ and significance level $\alpha \in (0,1)$, the confidence interval of the point estimate of the debaised LASSO estimator $\hat{\theta}^u$ is defined as:

$$\begin{aligned} J_i(\alpha) &\equiv \left[\hat{\theta}_i^u - \delta(\alpha,n), \hat{\theta}_i^u + \delta(\alpha,n)\right], \\ \delta(\alpha,n) &\equiv \Phi^{-1}(1-\alpha/2)\frac{\hat{\sigma}}{\sqrt{n}}[M\hat{\Sigma}M^\top]_{i,i}^{1/2} \end{aligned} \tag{26}$$

It can also be shown that the confidence interval is asymptotically valid, namely

$$\lim_{n\to\infty} \mathbb{P}\left(\theta_{0,i} \in J_i(\alpha)\right) = 1-\alpha \tag{27}$$

### 2.3.2 Hypothesis Testing

An important advantage of sparse linear regression methods such as the LASSO is their ability to provide a parsimonious explanation of the data, relying on only a small number of covariates. This can be simply understood as selecting the number of non-zero covariates $\hat{\theta}_i^n \neq 0$ for all $i \subseteq [p]$. However, the covariates estimated by the LASSO $\hat{\theta}_i^n$ do not provide a measure of the statistical significance for the finding that the coefficient is non-zero. In particular, we are interested in testing an individual null hypotheses $H_{0,i} : \theta_{0,i} = 0$ verses the alternative $H_{A,i} : \theta_{\theta,i} \neq 0$. Using the debiased estimator $\hat{\theta}^u$ derived in Algorithm 1, we construct the p-value $P_i$ for the test $H_{0,i}$ to be:

$$P_i = 2\left(1 - \Phi\left(\frac{\sqrt{n}|\hat{\theta}_i^u|}{\hat{\sigma}[M\Sigma\hat{M}^\top]_{i,i}^{1/2}}\right)\right) \tag{28}$$

The decision rule is based on the p-value $P_i$:

$$\widehat{T}_{i,\mathbf{X}}(y) = \begin{cases} 1 & \text{if } P_i \leq \alpha \quad (\text{reject } H_{0,i}), \\ 0 & \text{otherwise} \quad (\text{accept } H_{0,i}) \end{cases} \tag{29}$$

Where $\alpha$ is the fixed target Type I error probability. We measure the quality of the test $\widehat{T}_{i,\mathbf{X}}(y)$ in terms of its significance level $\alpha_i$ and statistical power $1-\beta_i$ where $\beta_i$ is the probability of type II error. It can be shown that this test achieves power $1-\beta = \alpha$. From a formal minimax perspective, given a family of tests $T_{i,\mathbf{X}} : \mathbb{R}^n \to \{0,1\}$ indexed by $i \in [p]$, $\mathbf{X} \in \mathbb{R}^{n\times p}$, we defined for $\gamma > 0$ a lower bound on the non-zero entries of $\theta_{0,i}$

$$\begin{aligned} \alpha_{i,n}(T) &\equiv \sup\left\{\mathbb{P}_{\theta_0}(T_{i,\mathbf{X}}(y) = 1) : \theta_0 \in \mathbb{R}^p, \|\theta_0\|_0 \leq s_0(n), \theta_{0,i} = 0\right\} \\ \beta_{i,n}(T;\gamma) &\equiv \sup\left\{\mathbb{P}_{\theta_0}(T_{i,\mathbf{X}}(y) = 0) : \theta_0 \in \mathbb{R}^p, \|\theta_0\|_0 \leq s_0(n), \theta_{0,i} \geq \gamma\right\} \end{aligned} \tag{30}$$

### 2.3.3 Generalizing to Simultaneous Confidence Intervals

In many situations, it is necessary to perform statistical inference on more than one of the parameters simultaneously. As an example, we might be interested in performing inference about $\theta_{0,R} \equiv (\theta_{0,i})_{i\in R}$ for some set $R \subseteq [p]$. The simplest generalization of the method method is when the number of indices $|R|$ stay finite as $n, p \to \infty$. The following Lemma is a generalization of Lemma 10.

**Lemma 11.** *Under the assumptions of Lemma 10, define*

$$Q^{(n)} \equiv \frac{\hat{\sigma}^2}{n}[M\hat{\Sigma}M^\top] \tag{31}$$

*Let $R = R(n)$ be a sequence of sets $R(n) \subseteq [p]$, with $|R(n)| = k$ fixed as $n, p \to \infty$ and further assume $s_0 = o(\sqrt{n}/\log p)$ with $s_0 \geq 1$. Then for all $x = (x_1, \ldots, x_k) \in \mathbb{R}^k$, we have*

$$\lim_{n\to\infty} \sup_{\theta_0 \in \mathbb{R}^p; \|\theta_0\|_0 \leq s_0} \left|\mathbb{P}\left\{\left(Q_{R,R}^{(n)}\right)^{-1/2}(\hat{\theta}_R^u - \theta_{0,R}) \leq x\right\} - \mathbf{\Phi}_k(x)\right| = 0 \tag{32}$$

*where $(a_1, \ldots, a_k) \leq (b_1, \ldots, b_k)$ indicates $a_1 \leq b_1, \ldots, a_k \leq b_k$, and $\mathbf{\Phi}_k(x) = \Phi(x_1) \times \cdots \times \Phi(x_k)$*

Mirroring the univariate inferential capabilities provided by Lemma 10, Lemma 11 provides the necessary theoretical results to move from marginal to simultaneous confidence regions. We can then define the valid confidence region $J_R(\alpha) \subseteq \mathbb{R}^k$ as

$$J_R(\alpha) \equiv \hat{\theta}_R^u + (Q_{R,R}^{(n)})^{1/2}\mathcal{C}_{k,\alpha} \tag{33}$$

Such that

$$\lim_{n\to\infty} \mathbb{P}\left(\theta_{0,R} \in J_R(\alpha)\right) = 1-\alpha \tag{34}$$

Where $\mathcal{C}_{k,\alpha} \subseteq \mathbb{R}^k$ is any Borel set such that $\int_{\mathcal{C}_{k,\alpha}} \phi(x)dx \geq 1-\alpha$ and $\phi(x) = \frac{1}{(2\pi)^{k/2}} \exp\left(-\frac{\|x\|^2}{2}\right)$. We can further construct a testing procedure that controls the family wise error rate (FWER) based on Bonferroni.

$$\widehat{T}^F_{i,\mathbf{X}}(y) = \begin{cases} 1 & \text{if } P_i \leq \alpha/p \quad (\text{reject } H_{0,i}) \\ 0 & \text{otherwise} \quad (\text{accept } H_{0,i}) \end{cases} \tag{35}$$

It then follows from the Bonferroni inequality that the asymptotic family-wise error rate is bounded above by $\alpha$:

$$\limsup_{n\to\infty} \text{FWER}(\widehat{T}^F, n) \leq \alpha \tag{36}$$

### 2.3.4 Non-Gaussian Noise

The inferential guarantees for the simultaneous confidence region $J_R(\alpha)$, the individual hypothesis tests, and the FWER-controlling procedure $\widehat{T}^F$ naturally extend to non-Gaussian noise, provided the independent errors $W_i$ have zero mean, variance $\sigma^2$, and a bounded moment $\mathbb{E}[|W_i|^{2+a}] \leq C\sigma^{2+a}$ for some $a > 0$. To ensure the asymptotic normality required for valid confidence intervals and $p$-values, the scaled summands of the error, denoted as $\xi_j = m_i^\top X_j W_j / (\sigma[m_i^\top \hat{\Sigma} m_i]^{1/2})$, must satisfy the Lindeberg condition. Namely, for every $\varepsilon > 0$, almost surely:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} \mathbb{E}\left(\xi_j^2 \mathbb{I}_{\{|\xi_j| > \varepsilon\sqrt{n}\}} \mid X\right) = 0 \tag{37}$$

Then $\sum_{j=1}^n \xi_k/\sqrt{n} \xrightarrow{d} N(0,1)$. This condition is satisfied by introducing an $\ell_\infty$ constraint to the debiasing optimization problem: $\|Xm_i\|_\infty \leq n^\beta$ for a fixed $\beta \in (1/4, 1/2)$. By bounding the maximum influence of any single observation, the Lindeberg condition holds, preserving the asymptotic Gaussianity that guarantees the validity of the intervals, the tests, and the FWER upper bound $\limsup_{n\to\infty} \text{FWER}(\widehat{T}^F, n) \leq \alpha$.

To construct the debiasing matrix $M = (m_1, \ldots, m_p)^T$ under the non-Gaussian regime, each row vector $m_i \in \mathbb{R}^p$ is obtained by solving the following convex optimization problem:

$$\begin{aligned} \underset{m\in\mathbb{R}^p}{\text{minimize}} \quad & m^T \hat{\Sigma} m \\ \text{subject to} \quad & \|\hat{\Sigma} m - e_i\|_\infty \leq \gamma, \\ & \|Xm\|_\infty \leq n^\beta \quad \text{for arbitrary fixed } 1/4 < \beta < 1/2 \end{aligned} \tag{38}$$

Consequently, under the assumptions of Theorem 8, the resulting p-values remain asymptotically valid in non-Gaussian settings; a formal statement and proof are omitted for brevity.

## 3 Empirical Analysis

To evaluate the theoretical results of Javanmard and Montanari (2014), we replicate six of the original simulation scenarios (selected due to computational constraints) and the real-data application on the riboflavin (vitamin $B_2$) dataset. Building on this baseline, we compare their method with the Lasso projection estimator (van de Geer et al., 2014; Zhang and Zhang, 2014; Bühlmann and van de Geer, 2015) (also known as the LASSO projection estimator), which was cited but not implemented in the original study. As the original work only benchmarked against multisample splitting and a ridge projection estimator, this additional comparison provides a more comprehensive assessment of its practical performance.

All computations were performed on a first-generation Intel NUC laptop equipped with a 12th Gen Intel Core i7-1260P processor. The replication study was conducted using `R` version 4.2.2, leveraging the `parallel`, `doParallel`, and `foreach` packages to enable parallel processing (R Core Team, 2024; Corporation and Weston, 2022; Microsoft and Weston, 2022). The implementation of the debiased LASSO proposed by Javanmard and Montanari (2014) was sourced from the authors' repository (`https://web.stanford.edu/ montanar/sslasso/`). Furthermore, routines for multisample splitting, the ridge projection estimator, and the LASSO projection estimator, alongside the riboflavin dataset, were accessed via the `hdi R` package (Ruben Dezeure et al., 2015).

### 3.1 Simulation Study

We consider the linear model (2) where rows of the design matrix $\mathbf{X}$ are fixed i.i.d. realizations from the $N(0, \Sigma)$, where $\Sigma \in \mathbb{R}^{p \times p}$ is a circulant symmetric matrix with entries $\Sigma_{jk}$ given as for $j \leq k$

$$\Sigma_{jk} = \begin{cases} 1 & \text{if } k = j \\ 0.1 & \text{if } k \in \{j+1, \ldots, j+5\} \\ & \text{or } k \in \{j+p-5, \ldots, j+p-1\} \\ 0 & \text{for all other } j \leq k. \end{cases} \tag{39}$$

Figure 1 illustrates an example of this matrix structure for $p = 600$. To generate the true regression coefficients, we select a uniformly random support set $S \subseteq [p]$ of size $|S| = s_0$. The non-zero coefficients are set to $\theta_{0,i} = b$ for $i \in S$, and $\theta_{0,i} = 0$ otherwise. Measurement errors are drawn as $W_i \sim N(0, 1)$ for $i \in [n]$. We evaluate several configurations of $(n, p, s_0, b)$, and for each configuration, the results are averaged over 20 independent realizations of the model with fixed design and regression coefficients. In contrast to Javanmard and Montanari (2014), who explicitly tune a regularization parameter and model optimization constraint $\mu$, we rely on the default configurations for all methods to better reflect how these implementations are typically used in practice.

For each configuration, 20 independent realizations of measurement noise and for each parameter $\theta_{0,i}$ the average length of the corresponding confidence interval is computed. Denoted by Avglength$(J_i(\alpha))$ where $J_i(\alpha)$ is the estimated confidence interval for $\hat{\theta}_i^u$ defined in Equation (26) and the average is taken over all realizations. The following metric considered is

$$\ell \equiv p^{-1} \sum_{i \in [p]} \text{Avglength}(J_i(\alpha))$$

Along with the average length of intervals for the active and inactive parameters as:

$$\ell_S \equiv s_0^{-1} \sum_{i \in S} \text{Avglength}(J_i(\alpha)), \quad \ell_S^c \equiv (p - s_0)^{-1} \sum_{i \in S^c} \text{Avglength}(J_i(\alpha)) \tag{40}$$

Similarly, the average coverage for the individual parameters is considered. i.e.

$$\widehat{\text{Cov}} = p^{-1} \sum_{i \in [p]} \widehat{\mathbb{P}}[\theta_{0,i} \in J_i(\alpha)], \tag{41}$$

$$\widehat{\text{Cov}}_S = s_0^{-1} \sum_{i \in S} \widehat{\mathbb{P}}[\theta_{0,i} \in J_i(\alpha)], \tag{42}$$

$$\widehat{\text{Cov}}_{S^c} = s_0^{-1} \sum_{i \in S^c} \widehat{\mathbb{P}}[\theta_{0,i} \in J_i(\alpha)] \tag{43}$$

Where probabilities ($\widehat{\mathbb{P}}$) are calculated from 20 realizations per configuration. Tables 1–4 compare interval coverage, false positive (FP) rates, and statistical power (TP) between this replication and Javanmard and Montanari (2014). Tables 2 and 4 show the relevant results listed in Javanmard and Montanari (2014).

As detailed in Tables 1 and 2, the replication produces tighter confidence intervals (e.g., $\ell \approx 0.15$ versus $\ell \approx 0.18$) and achieves a more precise alignment with the 95% nominal coverage compared to the original results, which tended toward over-coverage. When evaluating error rates (Tables 3 and 4), both studies successfully control FP rates near the nominal $\alpha = 0.05$ level. However, the replication achieves substantially higher statistical power (TP) across multiple methods in moderately sparse settings, most notably maintaining full power for multisample-splitting where the original study observed a significant drop.

Crucially, Table 3 highlights the performance of the LASSO projection estimator, which was not considered in the original study despite being cited. In direct comparison to the replicated Javanmard and Montanari method, the LASSO projection demonstrates superior sensitivity in low-signal regimes. For instance, under the $(1000, 600, 30, 0.1)$ configuration, the LASSO projection yields a TP of 0.8217 compared to the Javanmard and Montanari method's 0.7017, while still rigorously controlling the FP rate (0.0479 versus 0.0363). These findings indicate that the LASSO projecton estimator provides a more powerful framework for high-dimensional inference than the original Javanmard and Montanari procedure, improving signal detection without inflating Type I error.

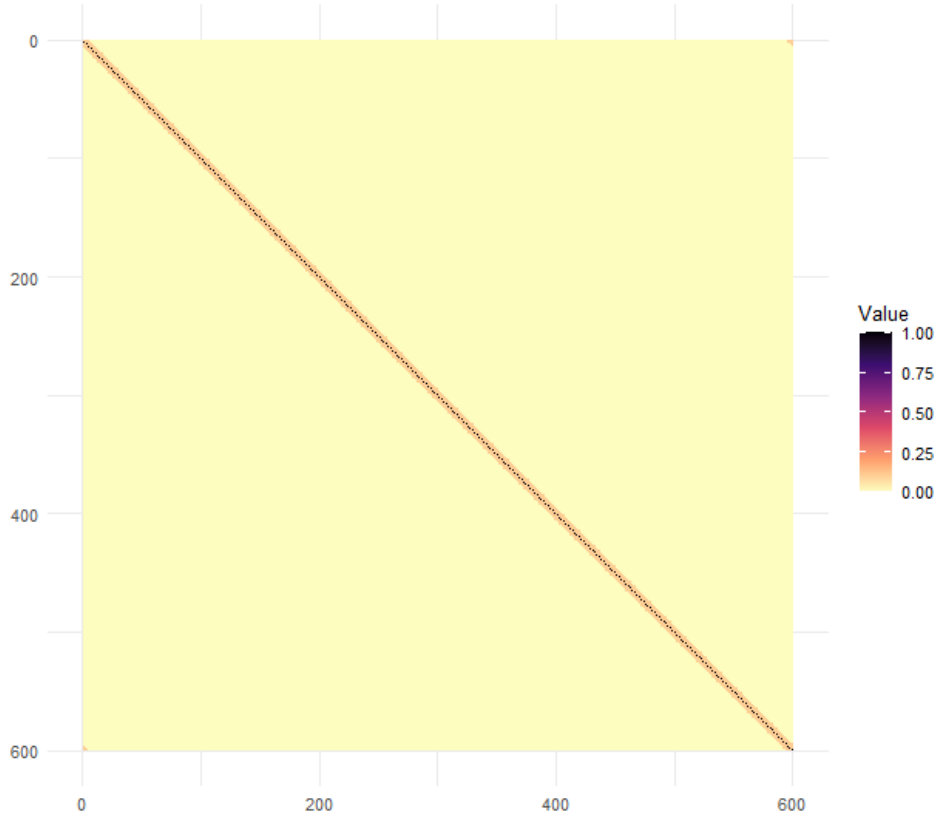


Figure 1: A visualization of the symmetric circulant matrix $\Sigma$ specified mathematically in Equation (39)

| Configuration \ Measure | $\ell$ | $\ell_S$ | $\ell_{S^c}$ | $\widehat{Cov}$ | $\widehat{Cov}_S$ | $\widehat{Cov}_{S^c}$ |
|---|---|---|---|---|---|---|
| (1000,600,10,0.5) | 0.1485 | 0.1485 | 0.1485 | 0.9592 | 0.9450 | 0.9594 |
| (1000,600,10,0.25) | 0.1473 | 0.1471 | 0.1473 | 0.9583 | 0.9600 | 0.9583 |
| (1000,600,10,0.1) | 0.1518 | 0.1521 | 0.1518 | 0.9602 | 0.9850 | 0.9597 |
| (1000,600,30,0.5) | 0.1528 | 0.1525 | 0.1528 | 0.9657 | 0.9617 | 0.9659 |
| (1000,600,30,0.25) | 0.1527 | 0.1528 | 0.1527 | 0.9680 | 0.9750 | 0.9676 |
| (1000,600,30,0.1) | 0.1590 | 0.1588 | 0.1590 | 0.9711 | 0.9600 | 0.9717 |

Table 1: Replicated results for synthetic data. Results correspond to 95% confidence intervals.

### 3.2 Data Analysis

To demonstrate our approach on empirical data, we analyze the riboflavin production dataset previously examined by Javanmard and Montanari (2014). Originally introduced by Bühlmann et al. (2014), the dataset comprises $n = 71$ samples and $p = 4,088$ covariates representing individual gene expression levels. The continuous response variable measures the log-transformed riboflavin production rate, which is modelled linearly against the log-transformed gene expression covariates.

Figure 2 shows that Multisample-splitting proved overly conservative with estimated confidence intervals which covered the whole range of the covariate distribution, whereas the debiased projection estimators and the Javanmard and Montanari (2014) approach successfully identified globally significant genes such as `ARGF_at` and `LYSC_at`. The Javanmard and Montanari method exhibited the highest precision , consistently yielding the narrowest confidence intervals and isolating signal where the Ridge and Lasso projection estimators were less sensitive. These results highlight the substantial precision gains achievable through the optimized de-biasing procedure proposed by Javanmard and Montanari (2014).

| Configuration \ Measure | $\ell$ | $\ell_S$ | $\ell_{S^c}$ | $\widehat{\text{Cov}}$ | $\widehat{\text{Cov}}_S$ | $\widehat{\text{Cov}}_{S^c}$ |
|---|---|---|---|---|---|---|
| (1000, 600, 10, 0.5) | 0.1870 | 0.1834 | 0.1870 | 0.9766 | 0.9600 | 0.9767 |
| (1000, 600, 10, 0.25) | 0.1757 | 0.1780 | 0.1757 | 0.9810 | 0.9000 | 0.9818 |
| (1000, 600, 10, 0.1) | 0.1809 | 0.1823 | 0.1809 | 0.9760 | 1 | 0.9757 |
| (1000, 600, 30, 0.5) | 0.2107 | 0.2108 | 0.2107 | 0.9780 | 0.9866 | 0.9777 |
| (1000, 600, 30, 0.25) | 0.1956 | 0.1961 | 0.1956 | 0.9660 | 0.9660 | 0.9659 |
| (1000, 600, 30, 0.1) | 0.2023 | 0.2043 | 0.2023 | 0.9720 | 0.9333 | 0.9732 |

Table 2: Simulation results for the synthetic data described in (Javanmard and Montanari, 2014). The results corresponds to 95% confidence intervals.

| | Javanmard & Montanari | | Multisample-splitting | | Ridge projection estimator | | LASSO projection estimator | |
|---|---|---|---|---|---|---|---|---|
| Configuration | FP | TP | FP | TP | FP | TP | FP | TP |
| (1000,600,10,0.5) | 0.0514 | 1 | 0 | 1 | 0.0420 | 1 | 0.0488 | 1 |
| (1000,600,10,0.25) | 0.0547 | 1 | 0 | 1 | 0.0385 | 1 | 0.0490 | 1 |
| (1000,600,10,0.1) | 0.0495 | 0.7700 | 0 | 0 | 0.0454 | 0.4750 | 0.0508 | 0.8550 |
| (1000,600,30,0.5) | 0.0531 | 1 | 0 | 1 | 0.0352 | 1 | 0.0418 | 1 |
| (1000,600,30,0.25) | 0.0518 | 1 | 0 | 0.9967 | 0.0413 | 0.9967 | 0.0433 | 1 |
| (1000,600,30,0.1) | 0.0363 | 0.7017 | 0 | 0.0200 | 0.0384 | 0.4383 | 0.0479 | 0.8217 |

Table 3: Replicated simulation results for false postive rates (FP) and and true positive rates computed at significance level $\alpha = 0.05$.

## 4 Discussion & Conclusion

The transition of high-dimensional regression from pure variable selection to formal statistical inference marks a critical methodological advancement. Because the non-linear $\ell_1$ penalty of the standard LASSO introduces inherent shrinkage bias, classical uncertainty quantification via confidence intervals and p-values was precluded. The framework proposed by Javanmard and Montanari (2014) successfully bridges this gap by constructing an optimal de-biasing matrix that corrects the LASSO estimate while explicitly minimizing asymptotic variance. The resulting theoretical guarantees for the estimator $\hat{\theta}^u$ related to its asymptotic normality, resilience to non-Gaussian noise, and adaptability for simultaneous testing provide a rigorous mathematical foundation for classical statistical inference in high-dimensional setting when $p \gg n$, under suitable sparsity assumptions.

Empirical evaluations across replications of the simulation study and the riboflavin dataset reveal a nuanced narrative regarding the practical performance of these estimators, highlighting a critical divergence between idealized settings and severely sample-starved, real-world applications. In the simulation study, multisample-splitting (Meinshausen et al., 2009) maintained strong statistical power, and the LASSO projection estimator (Zhang and Zhang, 2014) outperformed the Javanmard and Montanari (2014) method by demonstrating superior sensitivity in low-signal regimes. However, this dynamic reversed in a real high-dimensional genomic data setting. Furthermore, the Javanmard and Montanari estimator achieved the highest precision and successfully isolated empirical signals where the LASSO projection lacked sensitivity. This discrepancy suggests that while standard projection estimators excel under well-behaved, circulant covariance structures, the explicit convex optimization of the de-biasing matrix $M$ employed by Javanmard and Montanari (2014) adapts far more robustly to the complex correlation networks inherent in real biological data. While modern projection methods offer robust power in idealized scenarios, optimized de-biasing matrices produced by (Javanmard and Montanari, 2014) prove to be dominant for maximizing signal detection and enabling inference in noisy, high-dimensional settings.

To provide a more equitable evaluation of these estimators, it is recommended that future simulation studies employ design matrices that intentionally weaken or violate the restricted eigenvalue and compatibility conditions. The circulant covariance structure used in the simulation study features minimal off-diagonal correlation (as shown in Figure 1), creating an idealized setting where the LASSO projection estimator excels. Furthermore, these settings fail to capture the complex correlation structures inherent in real-world biological data. By benchmarking against highly correlated data generating processes, future research can more accurately demonstrate the practical necessity of the debaised estimator proposed by Javanmard and Montanari (2014).

| | Javanmard & Montanari | | Multisample-splitting | | Ridge projection estimator | |
|---|---|---|---|---|---|---|
| Configuration | FP | TP | FP | TP | FP | TP |
| (1000, 600, 10, 0.5) | 0.0452 | 1 | 0 | 1 | 0.0284 | 0.8531 |
| (1000, 600, 10, 0.25) | 0.0393 | 1 | 0 | 0.4 | 0.02691 | 0.7506 |
| (1000, 600, 10, 0.1) | 0.0383 | 0.8 | 0 | 0 | 0.2638 | 0.6523 |
| (1000, 600, 30, 0.5) | 0.0433 | 1 | 0 | 1 | 0.0263 | 0.8700 |
| (1000, 600, 30, 0.25) | 0.0525 | 1 | 0 | 0.4 | 0.2844 | 0.8403 |
| (1000, 600, 30, 0.1) | 0.0402 | 0.7330 | 0 | 0 | 0.2238 | 0.6180 |

Table 4: Simulation results for the synthetic data described in (Javanmard and Montanari, 2014). The false positive rates (FP) and the true positive rates (TP) are computed at significance level $\alpha = 0.05$.

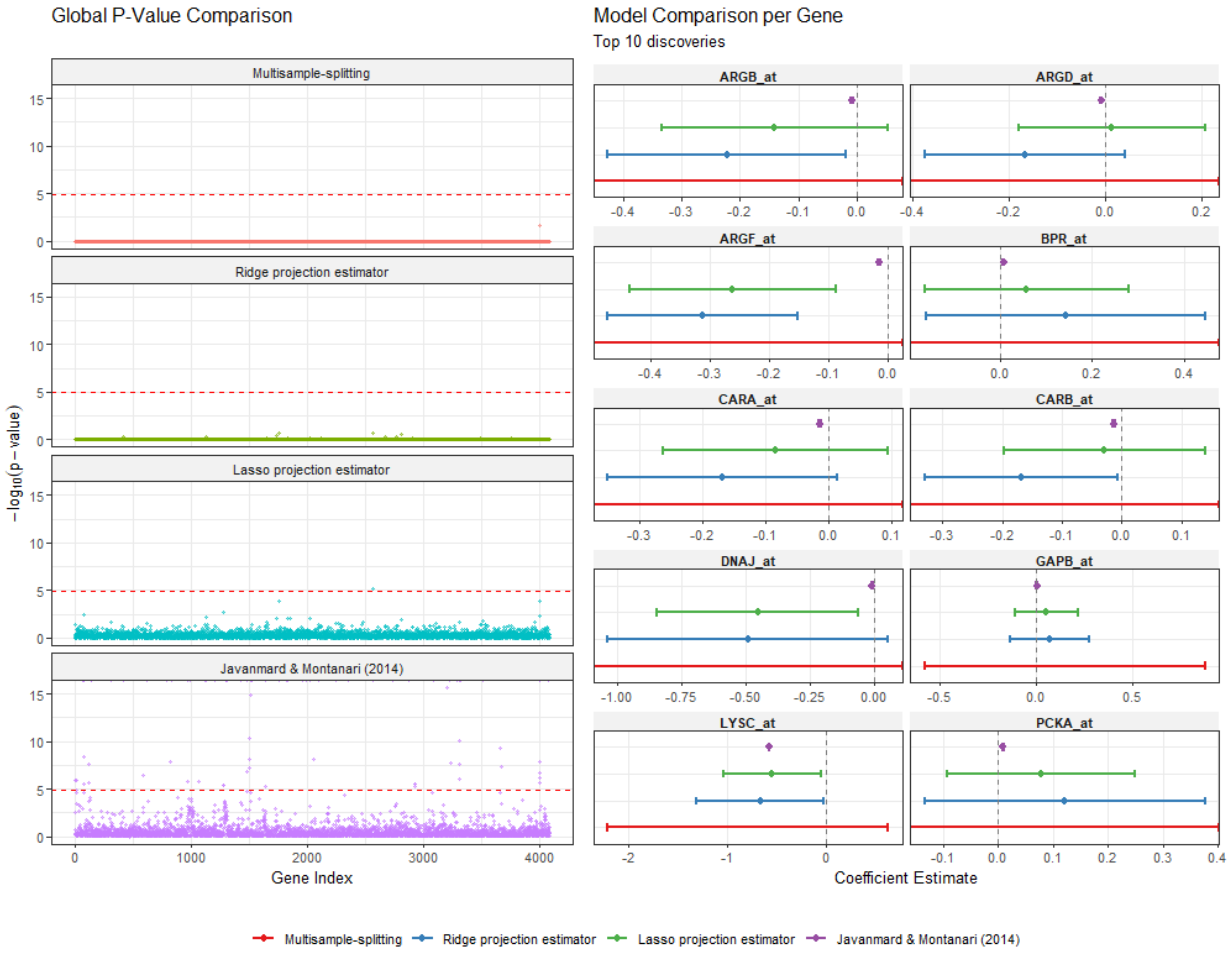


Figure 2: Comparative high-dimensional inference on the riboflavin dataset ($n = 71, p = 4,088$). The Manhattan plots (left) display the global p-value distribution relative to a Bonferroni-corrected threshold (red line), while faceted forest plots (right) provide 95% confidence intervals for the top 10 genes.

## References


Al-Ghuribi, S. M. and Noah, S. A. M. (2021). A comprehensive overview of recommender system and sentiment analysis.

Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *The Review of Economic Studies*, 81(2):608–650.

Belloni, A., Chernozhukov, V., and Wei, Y. (2013). Honest confidence regions for a regression parameter in logistic regression with a large number of controls. CeMMAP working papers 67/13, Institute for Fiscal Studies.

Bühlmann, P., Kalisch, M., and Meier, L. (2014). High-dimensional statistics with a view toward applications in biology. *Annual Review of Statistics and Its Application*, 1(1):255–278.

Bühlmann, P. and van de Geer, S. (2011). *Statistics for High-Dimensional Data: Methods, Theory and Applications*. Springer Series in Statistics. Springer Science & Business Media, Heidelberg.

Bühlmann, P. and van de Geer, S. (2015). High-dimensional inference in misspecified linear models. *Electronic Journal of Statistics*, 9(1):1449–1473.

Bühlmann, P. (2013). Statistical significance in high-dimensional linear models. *Bernoulli*, 19(4).

Chen, S. and Donoho, D. L. (1995). Examples of basis pursuit. In Laine, A. F. and Unser, M. A., editors, *Wavelet Applications in Signal and Image Processing III*, volume 2569, pages 564 – 574. International Society for Optics and Photonics, SPIE.

Corporation, M. and Weston, S. (2022). *doParallel: Foreach Parallel Adaptor for the 'parallel' Package*. R package version 1.0.17.

Fu, W. and Knight, K. (2000). Asymptotics for lasso-type estimators. *The Annals of Statistics*, 28(5).

Javanmard, A. and Montanari, A. (2013). Confidence intervals and hypothesis testing for high-dimensional statistical models. In Burges, C., Bottou, L., Welling, M., Ghahramani, Z., and Weinberger, K., editors, *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc.

Javanmard, A. and Montanari, A. (2014). Confidence intervals and hypothesis testing for high-dimensional regression. *Journal of Machine Learning Research*, 15(82):2869–2909.

Koren, Y., Bell, R., and Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37.

Kozak, S., Nagel, S., and Santosh, S. (2019). Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271–292.

Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*. Springer Texts in Statistics. Springer Science & Business Media, New York, NY, 3rd edition.

Lustig, M., Donoho, D., and Pauly, J. M. (2007). Sparse MRI: The application of compressed sensing for rapid MR imaging. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 58(6):1182–1195.

Meinshausen, N. and Bühlmann, P. (2010). Stability selection. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 72(4):417–473.

Meinshausen, N., Bühlmann, L. M., and Peter (2009). p -values for high-dimensional regression. *Journal of the American Statistical Association*, 104(488):1671–1681.

Meinshausen, N. and Bühlmann, P. (2010). Stability selection. *Journal of the Royal Statistical Society Series B (Statistical Methodology)*, 72(4):417–473.

Microsoft and Weston, S. (2022). *foreach: Provides Foreach Looping Construct*. R package version 1.5.2.

Minnier, J., Tian, L., and Cai, T. (2011). A perturbation method for inference on regularized regression estimates. *Journal of the American Statistical Association*, 106(496):1371–1382.

Peng, J., Zhu, J., Bergamaschi, A., Han, W., Noh, D.-Y., Pollack, J. R., and Wang, P. (2010). Regularized multivariate regression for identifying master predictors with application to integrative genomics study of breast cancer. *The Annals of Applied Statistics*, 4(1).

R Core Team (2024). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Ruben Dezeure, Peter Bühlmann, Lukas Meier, and Nicolai Meinshausen (2015). High-dimensional inference: Confidence intervals, p-values and R-software hdi. *Statistical Science*, 30(4):533–558.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B (Statistical Methodology)*, 58(1):267–288.

van de Geer, S., Bühlmann, P., Ritov, Y., and Dezeure, R. (2014). On asymptotically optimal confidence regions and tests for high-dimensional models. *The Annals of Statistics*, 42(3):1166–1202.

Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. Springer Texts in Statistics. Springer Science & Business Media, New York, NY.

Yamashita, O., aki Sato, M., Yoshioka, T., Tong, F., and Kamitani, Y. (2008). Sparse estimation automatically selects voxels relevant for the decoding of fmri activity patterns. *NeuroImage*, 42(4):1414–1429.

Zhang, C.-H. and Zhang, S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 76(1):217–242.